4.0L

Engine speed and crankshaft position are provided through the CKP sensor. The sensor generates pulses that are the input sent to the Powertrain Control Module (PCM). The PCM interprets the sensor input to determine the crankshaft position. The PCM then uses this position, along with other inputs, to determine injector sequence and ignition timing.

The sensor is a hall effect device combined with an internal magnet. It is also sensitive to steel within a certain distance from it.

The flywheel/drive plate has groups of four notches at its outer edge. On 4.0L 6-cylinder engines there are three sets of notches (Fig. 3).

The notches cause a pulse to be generated when they pass under the sensor. The pulses are the input to the PCM. For each engine revolution there are 3 groups of four pulses generated on 4.0L 6-cylinder engines.

The trailing edge of the fourth notch, which causes the pulse, is four degrees before top dead center (TDC) of the corresponding piston.

The engine will not operate if the PCM does not receive a CKP sensor input.

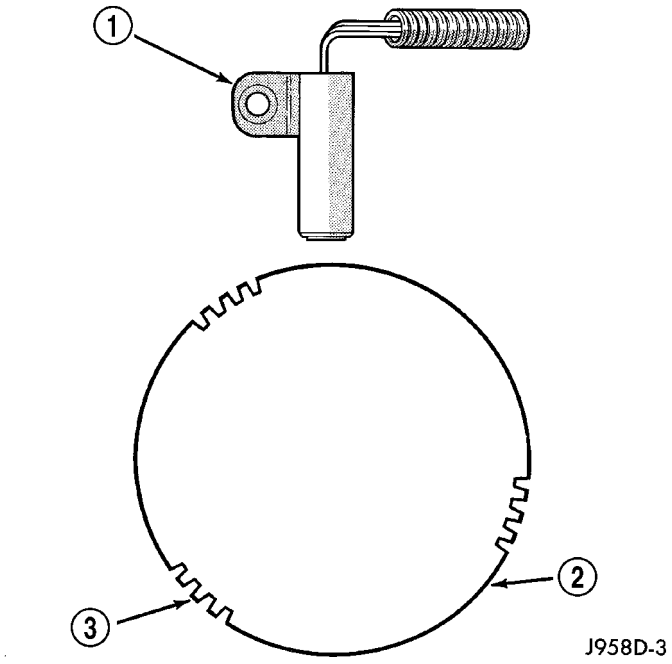

Fig. 3 CKP OPERATION - 4.0L

1 - CRANKSHAFT POSITION SENSOR
2 - FLYWHEEL
3 - FLYWHEEL NOTCHES

REMOVAL

2.4L

The Crankshaft Position (CKP) sensor is mounted into the right front side of the cylinder block (Fig. 4). It is positioned and bolted into a machined hole.

(1) Disconnect sensor electrical connector.
(2) Remove sensor bolt.
(3) Carefully twist sensor from cylinder block.
(4) Check condition of sensor o-ring (Fig. 5).

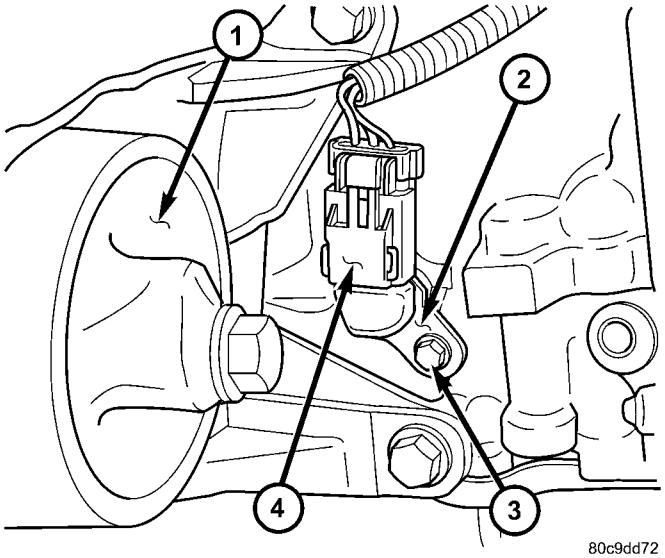

Fig. 4 CKP SENSOR LOCATION - 2.4L

1 - RIGHT FRONT ENGINE MOUNT
2 - CKP SENSOR
3 - MOUNTING BOLT
4 - ELECTRICAL CONNECTOR

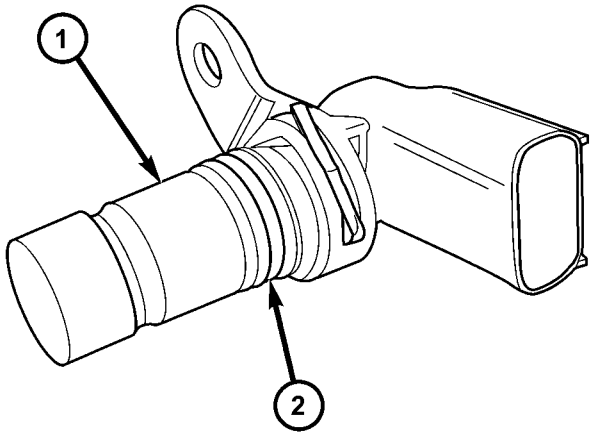

Fig. 5 CKP AND O-RING - 2.4L

1 - CKP SENSOR
2 - O-RING

4.0L

The crankshaft position (CKP) sensor is mounted to the transmission bellhousing near the rear of the engine block.

The sensor may be mounted to the transmission with one of the following four different configurations: